# Chapter 4 Consensus

Consensus lets replicas agree on one value even when some nodes fail or messages are delayed.
Safety means no two replicas commit different values; liveness means progress while a majority is alive.

## 4.1 Paxos

Paxos runs Prepare and Accept phases before a value is chosen.
A proposer first collects promises from a majority, then broadcasts the value it intends to commit.
It guarantees safety only; liveness relies on randomized retries and an extra leader election layer.
Paxos is famous for being hard to understand and hard to extend into a replicated log.

## 4.2 Raft

Raft splits the problem into leader election, log replication and safety.
A leader appends log entries and replicates them to followers before committing.
Randomized election timeouts and term numbers prevent split votes among candidates.
A candidate that collects votes from a majority becomes leader for that term.
The log matching property guarantees that two logs containing the same index and term are identical up to that point.
Membership changes are applied as special log entries so the configuration is itself replicated.
Snapshots compact the log and let a lagging follower catch up without replaying every entry.

## 4.3 ZAB

ZAB is the atomic broadcast protocol used by ZooKeeper.
It separates recovery mode from broadcast mode and orders committed transactions.
A new leader must synchronise its log with the previous leader before serving writes.
ZAB guarantees that committed transactions are delivered in the same order to every replica.
ZooKeeper builds locks, leader election and configuration management on top of this ordered log.